ORBIT OBSERVERS / CLEARSPACE / NEOSPACE PHASE 2

RPO Monte Carlo Simulation Handbook

A consolidated technical and project brief from our discussions: GEO servicing, tumbling clients, cooperative/degraded scenarios, Service Beacon, IOS mode, OSAN, HPC and the ESA OSIP-style roadmap.

Purpose Give the coding team a self-contained, defensible plan for building and testing the first RPO Monte Carlo simulator — while retaining the wider satellite-platform motivation.

One-minute reading guide

If you need…	Read…
Why this matters / what the product is	Sections 1–3: market, spacecraft states, novelty.
What to simulate and the equations	Sections 4–9: scenarios, physics, sensors, control, Monte Carlo.
How to code it and use MeluXina	Sections 10–13: architecture, verification, HPC.
How to connect results to the proposal	Sections 14–16: subsystem trades, OSAN, regulatory/business roadmap.
Original work to consult	Source notes and literature at the end.

The central engineering question

What minimum spacecraft-side preparation makes future GEO clients safely identifiable, approachable and recoverable — particularly when communications or attitude control have degraded?

Non-negotiable research integrity

- A Monte Carlo simulation can estimate **conditional performance under defined model assumptions**. It does not establish how frequently GEO satellites tumble in the real fleet.
- “Safe terminal approach” and “successful physical capture” are different outcomes. Do not claim capture success until alignment, relative speed, collision geometry and capture-interface limits are modelled.
- A Service Beacon is **not a new category of hardware by itself**; novelty must come from the fault-tolerant integration, measured value, interoperability and system requirements.

Prepared 15 October 2026 • Internal engineering working document • Literature references are supplied for verification.

1. Project history and platform vision

The ClearSpace challenge, **Roadmap for a Serviceable Satellite Platform**, asks a European spacecraft company to develop serviceability-by-design platforms supporting inspection, RPO, docking, refuelling, life extension and disposal — without converting every spacecraft into a fully modular repairable machine.

Why the problem exists

- Legacy GEO servicing can rely on mission-specific geometry: Northrop Grumman MEV used chemical apogee-engine nozzle / launch-adapter structures on compatible older clients. New hybrid/all-electric buses cannot be assumed to retain that nozzle.
- New GEO communications and weather satellites can remain functionally valuable while propellant, AOCS, a valve, power or avionics become life limiting; however, payload obsolescence can make simple life extension unattractive.
- Low-unit-value LEO constellations are often better served by inexpensive Design-for-Removal provisions than by expensive individual refuelling or hardware upgrades.
- A spacecraft is not “serviceable” merely because there is something to grip: servicer identification, consent, relative navigation, predictable safe/IOS modes, interface access and control authority are also needed.

The Phase-1 product-family logic

Tier	Minimum client preparation	Best-fit use
T0 – Removal-ready	MICE-like passive capture; fiducials/retroreflectors; serviceability data	LEO constellations, EOL/ADR
T1 – Cooperative-service	T0 + dedicated IOS mode and authenticated inter-vehicle information	Inspection, safer RPO; high-value missions
T2 – Life extension	T1 + refuelling/propulsion health and/or pod attachment	GEO telecom/weather
T3 – Augmentation	Power/data/thermal provisions for an externally attached technology or propulsion POD	Select high-value long-life clients

OSAN (Orbital Service & Augmentation Node) is our proposed multifunction service-zone integration concept, not an existing certified universal interface. **MSSI** describes the system-level mechanical, navigation, operational/data and EOL interface architecture.

Phase-2 shift

From vision to evidence Do not attempt to finish all seven Phase-1 work packages. Quantify the value of client preparation under nominal/degraded/tumbling conditions, then use those results to select the serviceability functions worth integrating into a future platform.

2. Which satellites are worth servicing?

Asset condition / orbit	Likely intervention	Economic interpretation
Healthy, fuel-limited GEO telecom	Refuel or external propulsion/AOCS extension	Strongest near-term life-extension rationale if payload still earns value.
High-value GEO weather / institutional	Inspect, selected life extension / anomaly management	High continuity value; stronger institutional mission-assurance constraints.
Operational but functionally ageing GEO	Software refresh or selective external upgrade POD	Upgrade must have functional paths to host bus/payload; cannot automatically replace internal OBC.
Degraded-but-recoverable GEO	Inspection, regain control, refuel/pod if conditions allow	Potential rescue opportunity, often time-critical and uncertain.
Completely dead or fragmented GEO	Relocation/ADR or debris-risk management	Rarely a commercial upgrade/life-extension client.
Mass-produced LEO constellation	Prepared capture and multi-client removal	Replacement normally preferable to expensive individual refurbishment.

Non-cooperative does not necessarily mean dead

- **Unprepared:** no dedicated navigation/capture aids, even if normal operations remain functional. A servicer may still recover it using advanced sensing/robotics.
- **Degraded cooperative:** useful spacecraft functions remain, but main OBC/comms or AOCS are partially unavailable; a separate service channel could retain limited state.
- **Dead but prepared:** main avionics/power are gone; passive fiducials and a structural capture interface still enable removal, but an active beacon cannot be relied upon indefinitely.
- **Uncontrolled tumbling:** a kinematic condition that can occur in some of the above states; not a separate measure of commercial value.

What the GEO evidence can and cannot say

Historic GEO anomaly studies have reported meaningful fractions of **anomaly events** involving attitude control, processors and propulsion. Those percentages are **not** fleet-wide probabilities that an active GEO spacecraft will tumble. Public GEO inventories do not establish a robust prevalence for the specific “degraded cooperative + tumbling” state.

Illustrative failures discussed: **Galaxy 15** (loss of command but recoverable), **Anik E2** (attitude-control recovery), **AMC-9** (drift/rotation and recovery for disposal), and **Intelsat 33e** (catastrophic fragmentation). These illustrate different outcomes, not an incidence distribution.

Implication RPO/tumbling Monte Carlo is a **resilience and recovery analysis**, not proof that tumbling is the largest paying GEO servicing market. Nominal refuelling/augmentation and EOL removal remain separate business cases.

3. Prior art: what is genuinely new?

The engineering literature already covers cooperative rendezvous, visual fiducials, proximity links, non-cooperative capture, health beacons, servicing modes, and Monte Carlo RPO simulations. The proposed innovation must therefore be specific and testable.

Established work	What our study adds (hypothesis)
Rendezvous beacons and proximity communications	A minimal independent, authenticated serviceability channel after main avionics degradation — if the value is measurable.
Safe mode / proposed servicing mode concepts	Explicit IOS state machine with command authority, inhibit logic and fault transitions for rendezvous and mated operations.
MICE and multifunction connectors such as MASI/ASSIST	Tiered client-side service zone allowing only the functions justified by mission value.
Non-cooperative RPO and Monte Carlo GNC	Paired, consistent comparison of unprepared, passive, beacon-assisted and actively cooperative client preparations.
Refuelling and propulsion pods	Health-aware choice among inspection, refuelling, POD, upgrade and retirement.

What the Monte Carlo can prove

- Under stated assumptions, how fiducials and/or time-tagged target state improve **navigation-state estimation** and predicted capture-point motion.
- How additional information changes **abort frequency, tracking accuracy, control effort and terminal feasibility**, once a closed-loop servicer model is implemented.
- Which communication performance (update rate, latency, outage tolerance) and client motion limits should become **preliminary platform requirements**.

What it cannot prove by itself

- Physical contact safety without detailed shape, contact dynamics, flexible appendages, attitude alignment and interface load verification.
- True commercial value without frequency of failures, customer willingness to pay, servicer scheduling/propellant, launch and qualification costs.
- Novelty of the beacon as an invention. An engineering study can demonstrate useful system integration, not patent novelty.

Best novelty statement “Quantifying minimum physical, behavioural and degraded-state information provisions needed to preserve safe servicing options across spacecraft failure states.”

4. Define the experiment before writing code

Primary research question

How much does client preparation improve RPO estimation and controlled terminal-approach feasibility when a GEO target rotates, communication degrades and servicer sensing is imperfect?

Architecture	Client provisions	What changes in model
C0 — Legacy / unprepared	No servicer-directed beacon/fiducials. Geometry estimated from generic recognition.	Lower pose reliability, incomplete capture metadata; vision-only observability.
C1 — Passively prepared	Known capture fixture geometry and fiducials; no active target packets.	Better pose measurement and known service-node reference.
C2 — Degraded cooperative	C1 + independent Service Beacon; ID/state/optional attitude rate; no stable AOCS assumed.	Time-tagged rate/health packets, outages, latency and sensor error.
C3 — Fully cooperative	C2 + functioning host AOCS, normal communication and IOS mode.	Attitude stabilisation/controlled behaviour and coordinated actions.

Crucial experimental fairness

- Compare C0, C1 and C2 first on the **same true tumbling trajectory**, target geometry and servicer performance. Only available measurements/metadata change.
- C3 genuinely changes target dynamics, so treat it as a separate **operational-mode benefit** — not as evidence that packets alone improved capture.
- Reuse paired random seeds and initial states across C0/C1/C2 so observed differences are attributable to client preparation. Include an ablation: C2-ID-only versus C2-with-rate.

Two progressively stronger claims

Stage I (minimal): “State-knowledge benefit” — errors and confidence intervals for prediction of moving capture-point position/velocity. This is feasible within days and is useful even without a complete controller.

Stage II (extended): “Closed-loop RPO benefit” — trajectories, terminal-approach and abort metrics with estimator/controller and keep-out constraints. Only call it capture feasibility if the physical contact conditions are modelled.

Deliverable A **serviceability envelope** that maps tumble, sensor quality and communication performance to a measurable readiness/safety outcome.

5. Reference GEO scenario and coordinate systems

A named spacecraft is **not needed** for the first study. A synthetic GEO client is often better for a general platform requirement study, with ClearSpace providing representative ranges where possible.

Group	Baseline data needed
Orbit	Reference circular GEO semimajor axis $a \approx 42{,}164$ km; Earth gravitational parameter $\mu \approx 3.986004418\times10^{14}$ m³/s²; $n = \sqrt{\mu/a^3} \approx 7.292\times10^{-5}$ rad/s.
Client bus	Approximate mass/box-wing geometry, inertias, capture location, exclusion zone, rotation model and control state.
Servicer	Relative start state, thrust/acceleration limit, mass if Δv→propellant is needed, reaction time and controller.
Navigation & ISB	Optical pose/rate accuracy, update rate, fields of view, dropouts; beacon rate accuracy, packet timing/loss/latency.
Servicing interface	Capture tolerances, contact normal/alignment, mechanical keep-out and permissible relative speed.

Frame definition prevents common mistakes

- **ECI:** inertial Earth-centred reference for orbital truth if using a nonlinear propagator.
- **LVLH / Hill (L):** rotates with the client reference orbit; x radial outward, y along-track, z cross-track (check chosen right-handed convention). Relative translation uses x,y,z in L.
- **Client body (B):** fixed to satellite structure; the capture location ρ_{cap}^{B} is fixed in this frame.
- **Rotation R_{BL}:** transforms client body coordinates into LVLH coordinates. Use one quaternion ordering/convention consistently.

The rotation rate used in the LVLH capture-point velocity is **body relative to LVLH**, not simply the inertial body rate. For short-duration toy simulations constant angular rate relative to LVLH is acceptable if explicitly stated.

Geometric assumptions for a coding prototype

Illustrative only (not ClearSpace requirements): client mass ≈ 3,500 kg; approximate bus 2.5 × 2.0 × 3.0 m; capture interface offset 1–2 m; close-range start 10–100 m; study target rotation 0–5 deg/s. Do not imply all 5 deg/s GEO cases are likely or capturable.

6. Orbital translation: equations and limits

Use the Clohessy–Wiltshire / Hill linearisation for relative motion about a near-circular target orbit. Chaser relative acceleration components a_x, a_y, a_z are control plus any modeled perturbation differences.

```
x_ddot = 3 n² x + 2 n y_dot + a_x

y_ddot = -2 n x_dot + a_y

z_ddot = -n² z + a_z

n = sqrt(mu/a³),   a_GEO ≈ 42164 km,   n_GEO ≈ 7.292e-5 rad/s
```

State: s = [x, y, z, vx, vy, vz]. Implement ds/dt = [vx, vy, vz, ax, ay, az]. Use RK4 or an adaptive ODE integrator rather than forward Euler for research runs.

Why not simulate orbital insertion or the whole GEO transfer?

The platform-preparation features influence the last proximity and capture phases most directly. From hundreds of metres to contact, relative attitude, optical measurements, interface geometry, thruster errors and control interlocks dominate the research question. Long-range GEO phasing and fleet logistics are separate workstreams.

When is CW inadequate?

- Long-range or long-duration motion where linearisation errors become significant; eccentric/inclined target orbit; substantial perturbations or mission-critical precision.
- Strong coupling with attitude effects, thruster plumes, flexibility or contact dynamics — these require different models beyond the basic CW propagation.
- Plan selected nonlinear two-body (or Basilisk/GMAT/Orekit) comparisons to establish where local CW errors become unacceptable for the chosen scenario.

One deterministic baseline

Place servicer at a known hold point with zero relative velocity, disable target tumble, disable noise and run a controlled approach. Check that position/velocity converge and no keep-out region is penetrated. Only then introduce rotation and measurements.

Physics verification With zero applied thrust, compare against an analytic CW state transition or a second trusted integrator; verify units and limiting behaviour. A plot that looks plausible is not validation.

7. Tumbling target and the moving capture point

Use a constant-rate 2-D or single-axis tumble for the first learning model. The 3-D upgrade uses a quaternion and the rigid-body Euler equation.

```
I omega_dot + omega × (I omega) = tau_ext

q_dot = (1/2) Omega(omega) q      (convention dependent)
```

For a torque-free rotating rigid body, body-axis angular velocity does **not** generally stay constant for arbitrary inertia and spin axis. Constant rate is only a deliberate first-order simplification.

Capture-point kinematics

```
r_cap^L = r_target^L + R_BL(q) rho_cap^B
```

```
v_cap^L = v_target^L + omega_(B/L)^L × [R_BL(q) rho_cap^B]
```

Relative to the target COM placed at the local origin, the first terms vanish in the target-centred model. In a rotating LVLH frame, use the body rotation rate relative to that frame in the second formula.

Why geometry dominates tumbling difficulty

A capture point at radius 1.5 m and 2 deg/s has tangential speed approximately rω = 1.5 × 0.0349 = **0.052 m/s**. This is already comparable to an illustrative 0.05 m/s contact-speed tolerance — without any translation of the target COM. These numbers are examples, not a universal capture requirement.

- A tumbler can present or hide an interface depending on solar-array/payload shadows, approach side and client attitude.
- The servicer may need to estimate the capture-point **orientation and normal direction**, not only its position; angular synchronisation may exceed actuator capability.
- A successful terminal state also needs controlled contact geometry, acceptable joint loads and physical accessibility — not just distance <0.2 m.

Practical scenario stages

Acquisition (hundreds of metres) → characterisation/hold → safe corridor selection → terminal tracking/interception → capture attempt or abort. Simulate only phases justified by the available sensor and actuation models.

8. Navigation, sensor fusion and Service Beacon

Vision-only channel

Use an explicit measurement function z_{vis} = h(s,q,geometry) + noise. A minimum proxy may output noisy line-of-sight, range and attitude/rate, but it must not give the controller the true capture point or true angular velocity.

- Unprepared (C0): model uncertain target shape/feature availability and higher pose-estimation noise/outages.
- Prepared (C1): fiducials and known interface coordinate system improve observable pose; specify the assumed improvement as a parameter, not a proven constant.
- Sensors may fail behind appendages, at unfavorable illumination, in eclipse or at certain target attitudes; include correlated visibility outages.

Independent Service Beacon (C2)

Model an optional low-power controller, independent communications chain, authenticated spacecraft/service ID and selected time-tagged state. The key test is whether independently measured angular rate adds meaningful information beyond vision.

```
z_omega,k = omega_true(t_k) + epsilon_k

gamma_k ~ Bernoulli(1 - p_loss)   [first-order model]
```

Packets arrive at t_k + latency; the filter should update using the measurement timestamp, not falsely treat old data as current. Add burst outages after the first independent-loss study.

Estimator hierarchy

- **Prototype:** a simple angle/rate tracking filter or alpha–beta model; assess 1–5 s capture-point prediction error.
- **Integrated simulation:** EKF/UKF or other state estimator tracking relative translation, target attitude and angular rate; explicitly record covariance and measurement innovations.
- **Important:** an ID-only beacon may help consent/identification but cannot directly improve rate estimation. Include ID-only and rate-enabled ablation cases.

Communications standards caveat

CCSDS Proximity-1 is a relevant architectural reference for short-range bidirectional sessions, but the currently published physical layer is specifically defined for UHF in the Mars environment. Do not claim it is a ready-made “1 m GEO IOS protocol.” Spectrum, link design and authentication are later requirements. [R5]

9. IOS mode, guidance, safety and docked control

Client-side state machine

Normal → Service Request → Authenticate → Service Standby → Approach Permitted → IOS Mode → Mated/Service → Verify → Release, with explicit reject/abort/safe-separation paths where operationally feasible.

- Before approach: verify operator consent/control authority, maintain agreed target attitude and provide navigation/interface data.
- During final approach: prevent unexpected thruster activity, wheel desaturation, solar-array motion or conventional FDIR transitions that could endanger the servicer.
- After latching: define a single authoritative attitude/orbit-control policy for the combined stack; the client cannot independently “fight” the servicer.
- After separation: ensure positive release and clearance before returning client propulsion/AOCS authority; define what happens if communications fail while mated.

Simple guidance for the prototype

```
a_cmd = -Kp (r_hat - r_ref) - Kd (v_hat - v_ref)

norm(a_cmd) <= a_max
```

Use reference hold points outside the body/appendage keep-out zone, then switch to a final capture-point tracking mode only when geometry and confidence allow. A naïve direct-line controller can fly through the satellite; it is not sufficient.

Outcome definitions

Outcome	Required logic
Safe hold/approach	Keep-out not violated, navigation confidence maintained, valid approach path and thrust bounds satisfied.
Abort/retry	Triggered by rate, relative speed, covariance, visibility, authentication, interface alignment or control-margin limits.
Terminal capture opportunity	Position, velocity and capture-normal alignment within assumed window for a defined dwell time; not physical latch verification.
Unsafe event	Body/appendage penetration, uncontrolled collision trajectory or failed safety-monitor action.

No unsafe-claim shortcut Report “terminal-condition pass rate” until interface compliance, contact forces and latch dynamics are explicitly simulated or experimentally tested.

10. Monte Carlo experiment design

Monte Carlo means dispersing uncertain inputs across many closed-loop runs. It does not replace a validated deterministic simulator. Separate **design variables** from **uncertain disturbances**.

Outer design sweep	Illustrative values / notes
Target tumble speed	0, 0.25, 0.5, 1, 2, 3, 5 deg/s; broad sensitivity envelope, not real fleet distribution.
Pose / position measurement quality	2, 5, 10, 25, 50 cm as proxy; choose phase-appropriate models and avoid applying 50 cm at contact without abort.
Beacon update rate	0.1, 0.5, 1, 2, 5 Hz; also “no beacon.”
Packet loss / outage	0, 0.1, 0.25, 0.5, 0.75; later burst duration model.
Latency	e.g., 0.1, 0.5, 1, 2 s; must use measurement timestamps.
Information content	ID-only; angular rate; attitude+rate; full IOS cooperation.

Randomised within every cell

- Initial target attitude, tumble axis and inertia uncertainty; initial servicer relative position/velocity error.
- Optical measurement noise and visibility, beacon sensor errors/timestamps/dropouts, thruster scale/misalignment/delay.
- Paired seeds and consistent truth trajectories for C0/C1/C2 to isolate the value of information.

Sampling strategy

Start with 25–100 test runs per small set of cells. Once debugging/verification passes, scale to 500–2,000 independent seeds per key cell. Use Latin hypercube or Sobol only if it helps cover a broader continuous uncertainty space; keep rare-event probabilities separate.

Warning about statistics A “0 collisions in 1,000 runs” result does **not** demonstrate flight-level collision risk is below 10^{-6}. Your simulation may miss important failure physics; uncertainty intervals and model inadequacy must be reported.

11. Metrics, uncertainty and publishable plots

Metric	What it means
P(terminal condition met)	Fraction meeting location, speed, alignment and safe-approach criteria under a specified scenario distribution.
P(abort) / P(unsafe)	Distinct outcomes; abort can be a successful safety response, not an engineering failure.
Navigation / capture-point error	RMS and 95th-percentile prediction error, including rate and time-tag handling.
Minimum clearance	Closest unintended approach to the body/appendage exclusion geometry.
Δv and mission time	Control effort/time; convert Δv to approximate servicer propellant using explicit mass/Isp assumptions.
Robustness boundary	Largest studied target-rate/noise/packet-loss combination meeting preselected conditional performance threshold.

Four essential figures

- **Plot 1:** terminal-condition pass rate against tumble rate for C0, C1, C2, C3 (noting C3 uses stabilised dynamics).
- **Plot 2:** C2 beacon ablation (ID-only vs gyro packets) against packet loss and update rate.

- **Plot 3:** 2-D servicing envelope — tumble rate × navigation noise, coloured by safe outcome fraction for selected client levels.
- **Plot 4:** trade curve — additional client power/mass/complexity versus reduction in rate-estimation error, abort burden or fuel/approach time.

Uncertainty intervals

```
p_hat = N_pass / N_runs

SE(p_hat) ≈ sqrt[p_hat (1-p_hat) / N_runs]
```

Use Wilson binomial 95% confidence intervals (or another documented method), especially for p near 0 or 1. The simple SE formula is explanatory only; it is unreliable near boundaries.

How to avoid a misleading result

- Do not tune controller parameters for one client and leave others under-tuned unless comparing this as a deliberate “same servicer design” decision.
- Do not silently reduce target angular rate in the Beacon-only case; only the cooperative IOS mode changes target control dynamics.
- Do not select sensor errors to guarantee a preferred solution. Include optimistic, nominal and pessimistic branches with parameter provenance.

12. Software architecture: what your coder should build

```
rpo_project/
  config/           # units, scenario variants, model assumptions
  dynamics.py       # CW and optional two-body propagation
  attitude.py       # q, omega, target capture-point geometry
  sensors.py        # camera/fiducial visibility & noisy observations
  beacon.py         # ID/state packets, latency, loss and timestamps
  estimator.py      # angle/rate estimator, later EKF
  guidance.py       # hold-point trajectory + feedback control
  safety.py         # exclusion zone, GO/NO-GO, abort/terminal tests
  simulate.py       # one independent deterministic/stochastic run
  montecarlo.py     # parameter sweeps, random seeds, parallelisation
  analyse.py        # confidence intervals, diagnostics, figures
  tests/            # exact limits, unit consistency, invariants
```

Minimum viable simulation algorithm

```
for scenario in fixed_parameter_grid:
    for seed in paired_seeds:
        truth = initialise_target_and_servicer(scenario, seed)
        estimate = initialise_estimator(prior_only=True)
        while not terminal:
            advance_target_and_chaser_truth(dt)
            measurements = sense_target_with_noise_and_occlusion()
            packets = beacon_model_with_timestamps_loss_latency()
            estimate = filter_update(measurements, packets)
            target_point_hat = predict_capture_point(estimate)
            command = guidance_and_safety(estimate, target_point_hat)
            apply_bounded_control(command)
            terminal = check_capture_window_or_abort_or_unsafe()
        save_one_row(scenario, seed, outcome, metrics)
```

Hard boundary: truth vs estimated state

The guidance controller may access only **estimated** states, not truth values such as the exact target angle, exact capture-point velocity or actual future packet receipt. Earlier simplified illustrations in this chat were learning prototypes and should not be reused as evidence without this separation.

Data schema

```
scenario,seed,tumble_deg_s,vision_sigma_m,beacon_enabled,
beacon_packet_loss,beacon_rate_hz,latency_s,theta0_rad,
status,terminal_pass,aborted,unsafe,minimum_clearance_m,
capture_point_error_m,relative_speed_mps,delta_v_mps,
elapsed_s,model_version,config_hash
```

Save one result row per run; optionally store trajectories for a small diagnostic subset only. This keeps HPC storage manageable and makes plots reproducible.

Companion script: `rpo_beacon_information_pilot.py` implements only Stage I (planar optical/rate prediction with a Kalman filter). Its output is a prediction-window proxy, not a closed-loop docking or collision result.

13. Verification and validation before HPC

A single believable trajectory is not sufficient. Verification must establish that the code solves the equations as written; validation must establish relevance to physical RPO (limited by available reference data).

Test	Expected behaviour
CW zero-control check	Compare deterministic zero-thrust output with analytic CW state-transition solution or high-accuracy reference.
Stationary, perfect sensors	Estimated pose/rate and true pose/rate agree; simple hold-point guidance converges without body penetration.
Zero beacon, identical noise	C2 with packets disabled must reduce to C1, to numerical tolerance.
Beacon ID-only vs rate data	ID-only packets must not magically improve angular-rate estimation.
Zero / all packets lost	Packet delivery model returns deterministic boundaries; no stale-data leakage.
Quaternion norm & frame check	||q|| remains unity; body-fixed capture point rotates consistently in LVLH.
Zero target rate	Moving capture-point term vanishes except reference-frame motion.
High angular rate / unreachable interface	Safety logic aborts; it does not force a success via unrealistic thrust.
Integrator/time-step check	Results converge as dt decreases; time and units are coherent.
Paired seeds	C0/C1/C2 use the same sampled true target and disturbance realisation.

Validation ladder

- **V0:** math/unit tests in Python; compare to analytical HCW and simple rotating-target geometry.
- **V1:** synthetic observations + estimator error statistics; demonstrate no truth leakage.
- **V2:** closed-loop safe approach, hold points, aborts; inspect individual trajectories and failure logs.
- **V3:** Monte Carlo paired trials; convergence of metrics and confidence intervals.
- **V4:** cross-check selected cases in Basilisk or another trusted astrodynamics environment; later HIL / docking rig.

Model maturity label “Early-stage, scenario-based, low-to-medium-fidelity RPO robustness assessment”, unless and until representative interfaces, sensor models and capture dynamics are validated with the company.

14. MeluXina HPC workflow

Your laptop can solve one CW trajectory. MeluXina is useful because thousands of independent combinations must be analysed, reproduced and statistically compared. This is an embarrassingly parallel CPU workload.

Suggested workflow

- Build and test one complete scenario locally; profile runtime and memory; limit outputs to compact tables and selected traces.

- Move code to MeluXina using the access methods from LuxProvide onboarding; confirm allocation, account, compute partition and storage location.
- Use Slurm job arrays or bounded multi-process workers. Partition by scenario cells, not by rendering huge images or making every integrator step a separate job.
- Start with a small pilot job and validate outputs, then run the full Monte Carlo; version-control configuration and save reproducible seeds.
- Aggregate results locally or on an approved interactive/analysis node; export figures and summary tables before the allocation closes.

Illustrative Slurm job array (adapt to allocation)

```
#!/bin/bash
#SBATCH --job-name=GEO_RPO
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=16
#SBATCH --array=0-31
#SBATCH --time=00:45:00
# Account, partition and environment depend on LuxProvide guidance.
python montecarlo.py --chunk ${SLURM_ARRAY_TASK_ID} --out results
```

Do not assume MeluXina always grants a whole node or the same modules/partitions across projects. Confirm site instructions and resource policy during onboarding.

What to request

Our earlier rough allocation estimate was **100 GB initially** for numerical RPO sweeps and result processing, with **200–300 GB** only if substantial synthetic image/ML datasets are introduced. CPU parallelism is the main requirement; GPU only if actual optical-image inference/training becomes part of scope.

When will HPC really matter?

First ~100 local runs test correctness. A few thousand cases test convergence. For a large grid with hundreds or thousands of samples per cell, parallel HPC allows robust sensitivity maps within the limited Phase-2 schedule.

15. Service Beacon feasibility and GEO detumbling

Independent Beacon: resilience, not immortality

A beacon need not operate entirely from a standalone battery for a full 15–20-year GEO mission. A plausible architecture uses the normal spacecraft bus in nominal life and a **qualified independent reserve** for a defined degraded-state survival window.

```
E_usable (Wh) >= P_avg (W) × T_survival (h)
```

```
P_avg ≈ P_sleep + duty_TX × P_TX + duty_sensor × P_sensor
```

Example only: 0.01 W sleep + 1% duty at 2 W TX + 10% duty at 0.03 W sensing ≈ 0.033 W. Thirty days ≈ 23.8 Wh **ideal** load energy; usable battery capacity must allow for aging, temperature, conversion losses, self-discharge, radiation and end-of-life margin.

- Size the emergency survival period based on a realistic rescue response plan, not an arbitrary “works indefinitely” assertion.
- Study antenna coverage under tumble, long dormancy reliability, wake-on-command vs periodic pulses, authentication and anti-spoofing.
- After complete power failure, fall back to **passive** fiducials, retroreflectors, known capture geometry and prepared EOL provisions.

Can a GEO target be passively detumbled?

- Gravity-gradient torque at GEO is weak and is **not inherently dissipative**; it cannot be assumed to stop tumble by itself.

- Conventional Earth-field magnetorquer torque weakens strongly at high altitude; do not extrapolate a LEO detumbling solution to GEO.
- Servicer-generated local magnetic/eddy-current, electrostatic or robotic/contact damping are research directions; require power, geometry, time and safety/feasibility studies.
- An optional parametric sub-study may model an abstract detumbling law, but it must **not** claim a physical detumbler design unless torque authority is separately substantiated.

Important focus decision Keep beacon survival and detumbling as **architecture sensitivity branches**. The primary Phase-2 deliverable remains quantifying navigation/control benefit of preparation.

16. OSAN, refuelling, pods and fleet-level economics

OSAN is an integration study, not a new docking mechanism

A future GEO service zone can bundle passive capture/load transfer, approach/fiducial references, IOS data, optional power/data/fluid connections and accommodation for an external augmentation POD. Compare legacy chemical, hybrid and all-electric buses; do not assume one identical coupling or fuel standard fits all.

Servicing choice	When attractive	Technical condition
MEV-style attached propulsion	Legacy unprepared client with functional payload but limited AOCS/propellant	Compatible mechanical capture; servicer committed to attached client.
Refuel-and-release	Prepared propulsion system remains healthy and compatible; repeated client visits feasible	Valves/tanks/feed health + qualified fluid connector + transfer verification.
Propulsion POD	Host still valuable but original propulsion is doubtful or cannot be refuelled	Structural attachment + coupled orbit/attitude control + adequate mass properties.
Upgrade POD	Host physically and operationally useful but selected functionality ageing	Power/data/thermal/software interfaces available; no assumption that old OBC is magically replaced.
Removal	Low recovery value or non-operational client	Prepared capture and authorised disposal strategy.

Servicer itself has finite resources

One reusable servicer cannot visit an unlimited number of GEO clients. Transfers consume its manoeuvre propellant; refuelling consumes delivered propellant, POD deployment consumes cargo and launch mass; service cycles cost time and risk.

```
Delta-v = Isp * g0 * ln(m_before / m_after)

N_visits constrained by maneuver fuel + delivered inventory + time + risk reserve
```

- MEV-like attached extensions reduce throughput while attached; refuel-and-release and detachable pods may increase visits but need replenishment and client qualification.
- A second, optional business-simulation branch could optimise GEO client sequencing and resource utilisation; do not make it the main RPO coding priority until Phase-2 safety evidence is secure.

Design-for-value rule

Compare the **discounted incremental benefit** of servicing against interface mass/launch/integration/qualification, servicing price, operational downtime and risk. Avoid double counting retained revenue and avoided replacement costs.

17. Regulatory, business and ESA OSIP link

Regulatory and standardisation direction

- **RPO safety:** ISO 24330:2022 and relevant ESA close-proximity guidance frame responsible operations and information sharing; formal licensing is mission- and State-specific.
- **Space debris / EOL:** ISO 24113, ESA debris-mitigation/Design-for-Removal work, IADC principles; preparedness does not itself guarantee a successful disposal mission.
- **Interfaces:** ECSS-E-ST-10-24C Rev.1 supports formal interface management and controlled ICDs. MICE/MASI/ASSIST are engineering references, not a unified certified interface chosen by our study.
- **Communication:** CCSDS inter-vehicle concepts are references; interoperable authenticated GEO servicing communications remain a design/standards selection task.
- **Owner/operator consent and national authorisation:** define command authority, contractual responsibilities, records and service plan. The EU Space Act direction is a proposed/evolving driver, not already settled law.

Who might pay?

Commercial GEO telecom operators, high-value institutional/weather owners, spacecraft OEMs and servicing providers are the likely early stakeholders. LEO fleet operators are potential customers for low-mass Design-for-Removal and multi-target disposal. Future rescue/removal subscriptions and “launch lighter, refuel later” are **business hypotheses** requiring economics and service availability.

How OSIP-style evaluators should see the idea

Evaluation thread	Evidence to supply
Novelty/state of art	Compare known fiducials, health beacons, IOS modes, docking and RPO with your precise degraded-cooperative integration gap.
Technical merit	Verified relative dynamics + information-value Monte Carlo + IOS authority/safety architecture.
Expected results	Serviceability envelope, minimum ISB fields/rate, safe-mode/IOS state machine and requirements trace.
Implementation	Software-in-loop → ground communication/controller demo → HIL/contact testing → IOD → product catalogue.
Industrial value	Tier-based mass/complexity trade, commercial target cases, operator/regulatory adoption gates.

18. Practical work split, calendar and acceptance criteria

Team lead	Main responsibility	Definition of done
1. Systems/OSIP	Requirements, common assumptions, proposal, ClearSpace coordination	One consistent scope, scenario list, risk/TRL/roadmap.
2. Relative dynamics	CW, target tumble, capture geometry, guidance	Validated single-case RPO simulations and test suite.
3. HPC/stats	Monte Carlo, paired seeds, sampling and MeluXina	Reproducible envelopes with CIs and diagnostics.
4. Beacon/comms	Independent channel concept, link loss/latency, power budget	Data fields, mode/packet model and survival trade.
5. IOS/docked + OSAN	Safe state machine, command authority, generic service-node layout	CONOPS + hazard/abort and interface allocation.
6. Economics/regulation	Tier-to-client map, standards, servicing vs replacement	Market assumptions, burden/value trade and adoption plan.

Priority order for the coding team

- Day 1: frame conventions, CW propagation, inertial-to-LVLH tests, stationary capture-point geometry.
- Day 2: rotating target/capture point; validation of rate and transformed velocity.
- Day 3: true-vs-estimated separation, vision/fiducial observations; C0 and C1.
- Day 4: C2 beacon timestamped rate packets, dropouts, error, rate prediction; IOS logic for C3.
- Day 5: controller/hold/abort and first paired-seed local Monte Carlo; verify consistency and safety.
- Day 6: MeluXina sweeps, confidence intervals, sensitivity figures and anomaly case analysis.
- Day 7: annotate model limits, derive requirements, freeze OSIP proposal plots and team presentation.

Minimum credible result versus stretch goal

Minimum: validated CW + tumble/capture geometry + estimated-state predictor + paired Monte Carlo showing capture-point prediction errors for C0/C1/C2. This is a valid information-value study, even before end-to-end capture.

Stretch: integrated, bounded closed-loop final approach with explicit keep-out, align/velocity, abort and Delta-v metrics. Include C3 cooperative mode and additional detumbling sensitivity only after verification.

What success sounds like “For a representative GEO client and explicitly stated uncertainties, we identified the failure/rotation regimes where passive preparation, independent state data and active IOS cooperation materially change servicing feasibility — and translated those results into requirements and a staged product roadmap.”

Research and technical references

These links provide a starting reading set. The main body distinguishes previously discussed concepts and engineering assumptions from verified source facts; verify all numbers and legal applicability before external submission.

R1. Barbee et al. (2010), Guidance and Navigation for Rendezvous and Proximity Operations with a Non-Cooperative Spacecraft at Geosynchronous Orbit (NASA NTRS).
https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20100019266.pdf

R2. Guidance and Control Architecture for Rendezvous and Approach to a Non-Cooperative Tumbling Target, Aerospace (2025).
https://www.mdpi.com/2226-4310/12/8/708

R3. Basilisk official learning documentation: spacecraft simulation, integrated scenarios and Monte Carlo examples.
https://avslab.github.io/basilisk/Learn.html

R4. Basilisk examples index: Monte Carlo and proximity/nav demonstrations.
https://avslab.github.io/basilisk/examples/index.html

R5. CCSDS Proximity-1 physical layer; short-range links, currently UHF Mars physical-layer scope.
https://ccsds.org/Pubs/211x1b4e1.pdf

R6. ISO 24330:2022, RPO/OOS programmatic principles and practices.
https://www.iso.org/standard/78463.html

R7. Basilisk source repository and reproducible scenario examples.
https://github.com/AVSLab/basilisk

Questions to take back to ClearSpace

- What representative GEO target inertias, rotation-rate envelopes, capture-point geometry, hold-point distances and interface tolerances can they share?
- Which client state is the more valuable study: operational but unprepared, degraded/partially controllable, or fully dead EOL?
- Would they value a minimum independent angular-rate estimate, or is identity/status/command coordination more operationally important?
- What standard RPO/abort metrics do they use, and how would they define a “safe serviceable envelope” for an early study?
- Does the first target product need EOL recovery or prepared GEO life extension? Which interface functions add unacceptable client mass/qualification burden?

Final checklist before publishing results

☐ Frames/units documented ☐ controller uses estimates only ☐ scenario assumptions logged ☐ paired seeds & sampling recorded ☐ abort ≠ collision ≠ capture ☐ confidence intervals shown ☐ literature numbers checked ☐ no claim of flight qualification.